

THE POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY 1



BUSSINESS INTELLIGENT REPORT

Topic: Analyzes e-commerce data.

CLASS: E22HTTT

GROUP : 6

NAME: NGUYỄN TUẤN MINH - B22DCKH077
NGUYỄN XUÂN KIÊN - B22DCKH062

Hà Nội - 2026

TABLE OF CONTENTS

PART I: INTRODUCTION & BUSINESS CONTEXT.....	2
1.1. Market Context and the Importance of Business Intelligence in E-Commerce.....	2
1.2. Project Rationale and Objectives.....	2
1.3. Target Audience and Stakeholders.....	3
PART II: DATASET DESCRIPTION & PREPROCESSING.....	4
2.1. Overview of the Olist E-Commerce Dataset.....	4
2.2. Relational Schema of Staging Tables.....	4
2.3. Description of Key Data Entities.....	5
2.4. Data Quality Challenges & Preprocessing Requirements.....	6
PART III: DATA WAREHOUSE DESIGN.....	7
3.1. Analytical Model Selection: Star Schema.....	7
3.2. Data Warehouse Schema Diagram (ERD).....	8
3.3. Table Architecture and Data Dictionaries.....	8
PART IV: ETL PROCESS (EXTRACT, TRANSFORM, LOAD).....	12
4.1. ETL Architecture and Pipeline Flow.....	12
4.2. Extract Phase (E): Raw to Staging.....	14
4.3. Transform Phase (T): Staging to Star Schema.....	14
2. English Category Translation (Dim_Product).....	15
3. Deduplication and Cardinality Reduction (Fact_Sales).....	16
4.4. Load Phase (L): Populating and Constraints.....	17
PART V: DASHBOARD IMPLEMENTATION (BI DASHBOARD).....	20
5.1. Tool Selection and Data Connection.....	20
5.2. DAX Measure Development.....	21
5.3. Dashboard Pages and Visual Layouts.....	22
PART VI: INSIGHT ANALYSIS.....	30
6.1. Sales Revenue & Seasonal Trends.....	30
6.2. Consumer Shopping Behavior & Payment Preferences.....	31
6.3. Product Category Performance.....	34
6.4. Geographic Distribution & Regional Market Concentration.....	35
6.5. Logistics Efficiency & Shipping Cost Burden.....	37
6.6. Customer Satisfaction & Service Quality Correlation.....	40
PART VII: CONCLUSION.....	42
7.1. Project Summary.....	42

7.2. Key Strategic Takeaways.....	43
7.3. Future Scope and Enhancements.....	43

PART I: INTRODUCTION & BUSINESS CONTEXT

1.1. Market Context and the Importance of Business Intelligence in E-Commerce

Over the past decade, Electronic Commerce has revolutionized the global retail landscape, transforming how consumers interact with brands and purchase goods. In emerging economies, E-commerce has witnessed exponential growth, driven by increasing internet penetration, smartphone adoption, and a shifting consumer preference towards the convenience of online shopping.

However, this rapid digital transition has generated an unprecedented volume of transactional, logistics, and customer feedback data. For e-commerce platforms and marketplace integrators, managing and extracting value from these complex datasets is a significant challenge. Raw transactional databases are typically optimized for Online Transaction Processing (OLTP) and are not suited for analytical queries.

This is where Business Intelligence (BI) becomes critical. By implementing a structured Data Warehouse (DW) using dimensional modeling and establishing automated Extract, Transform, and Load (ETL) pipelines, businesses can consolidate disparate data sources. BI systems transform raw operational records into clean, historical, and multidimensional data structures. Consequently, this enables organizations to perform complex analytical queries, design interactive dashboards, and make data-driven decisions that enhance operational efficiency, minimize costs, and maximize revenue.

1.2. Project Rationale and Objectives

This project focuses on analyzing the dataset provided by Olist, the largest department store integrator in Brazilian marketplaces. Olist operates as a platform that connects small and medium-sized merchants directly to Brazil's main e-commerce portals, handling the cataloging, logistics, and customer communication on behalf of the sellers.

The Olist dataset represents a rich, real-world e-commerce environment containing approximately 100,000 orders from 2016 to 2018. It details various aspects of the transaction lifecycle, including order progress, payment terms, item dimensions, customer

locations, and post-purchase customer reviews. Analyzing this dataset provides valuable insights into the multi-faceted dynamics of a large-scale marketplace.

The main objectives of this Business Intelligence project are:

1. **Sales and Revenue Performance Analysis:** Identify seasonal patterns, calculate monthly revenue trends (Month-over-Month growth), and determine which product categories drive the highest sales volume and profit margins.
2. **Logistics and Shipping Optimization:** Evaluate shipping efficiency by comparing estimated delivery dates with actual delivery times. Additionally, analyze the distribution of freight costs across different geographic states to identify regions with high delivery overheads.
3. **Customer Experience and Service Quality Assessment:** Investigate the relationship between customer satisfaction (measured through review scores) and operational metrics, such as delivery delays and payment installments, to identify key pain points in the customer journey.

1.3. Target Audience and Stakeholders

The insights generated by the BI dashboards are designed to serve three primary groups of stakeholders within the organization:

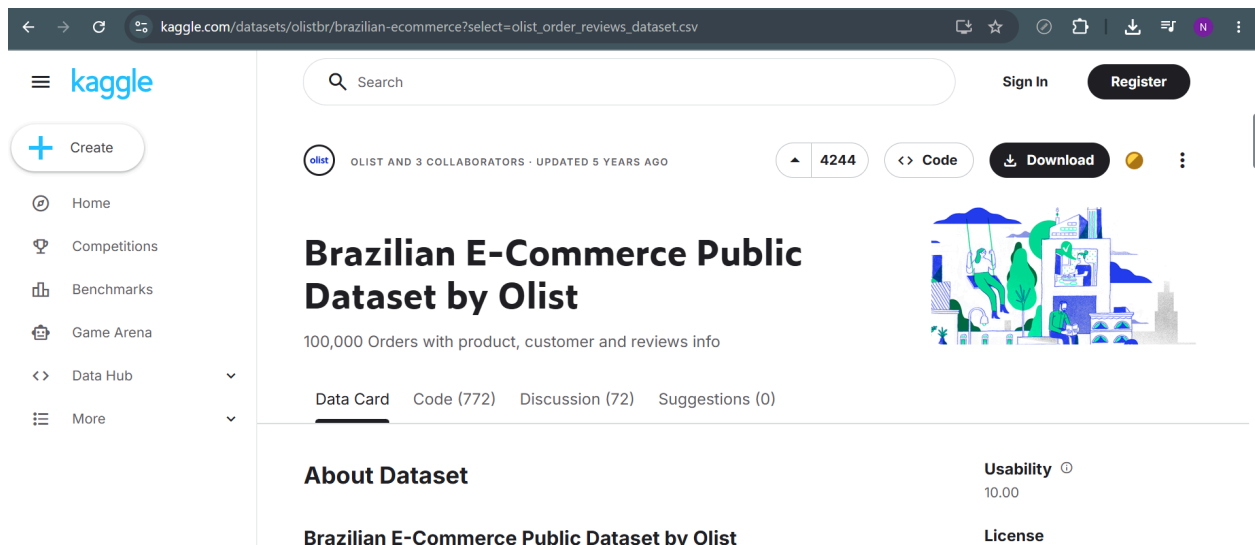
- **Executive Leadership (Board of Directors):** They require high-level, aggregated key performance indicators (KPIs) such as total revenue, total orders, and overall market growth trends to guide long-term business strategy, investment distribution, and market expansion plans.
- **Operations and Logistics Managers:** They require detailed geographical and timeline analysis to monitor logistics performance, evaluate carrier efficiency, and implement strategies to reduce freight costs and delivery times in remote regions.
- **Marketing and Customer Relationship Management (CRM) Teams:** They leverage demographic insights and product performance data to design targeted marketing campaigns, optimize the product catalog, and address quality issues in product categories that receive low customer ratings.

PART II: DATASET DESCRIPTION & PREPROCESSING

2.1. Overview of the Olist E-Commerce Dataset

The foundation of this Business Intelligence project is the Olist Brazilian E-Commerce Dataset, which contains real commercial transactions provided by Olist. The dataset covers a total of 99,441 orders placed between September 2016 and September 2018 across multiple marketplaces in Brazil.

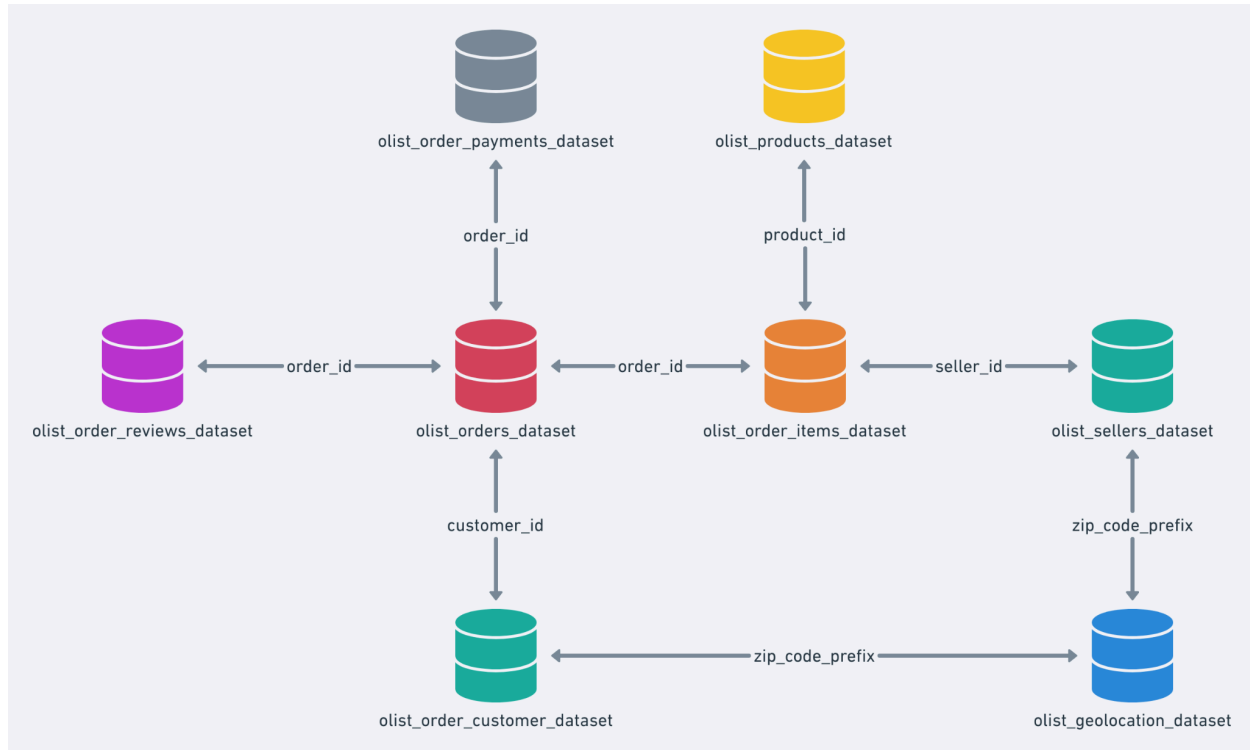
This dataset provides a comprehensive, multi-dimensional view of typical e-commerce operations. It captures the customer journey from the initial order placement, through payment confirmation, seller processing, shipping logistics, and finally to the post-delivery customer review. The data has been anonymized to protect customer and seller privacy, but it maintains structural and statistical integrity.



Dataset website picture

2.2. Relational Schema of Staging Tables

Before constructing the Data Warehouse, the raw dataset is imported directly into a staging database (Staging Area) to ensure data persistence and allow initial inspection. The staging area consists of 9 distinct tables (excluding geolocation data) that exhibit a relational structure where orders serve as the central hub.



Relational Database Schema of Olist Staging Area.

2.3. Description of Key Data Entities

Below is a detailed description of the staging tables used to build the analytical model:

1. **stg_customers:** Contains demographic information about the buyers.
 - **customer_id:** Unique identifier linking each customer record to a specific order.
 - **customer_unique_id:** Persistent identifier for each customer (used to track repeat buyers).
 - **customer_zip_code_prefix, customer_city, customer_state:** Location details of the customer.
2. **stg_orders:** Records the state and timeline milestones of each order.
 - **order_id:** Unique identifier of the transaction.
 - **customer_id:** Foreign key referencing the customer.
 - **order_status:** Status of the order (e.g., delivered, canceled, shipped).
 - **Timestamps:** order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date.

3. **stg_order_items:** Contains the itemized details of products purchased within each order.
 - **order_id:** Foreign key referencing the order.
 - **order_item_id:** Sequential number of items within the same order.
 - **product_id:** Foreign key referencing the product.
 - **seller_id:** Foreign key referencing the merchant.
 - **price:** Selling price of the product.
 - **freight_value:** Shipping cost charged to the customer.
4. **stg_products:** Details product characteristics.
 - **product_id:** Unique identifier of the product.
 - **product_category_name:** Native category name in Portuguese.
 - **Specifications:** product_weight_g, product_length_cm, product_height_cm, product_width_cm.
5. **stg_payments:** Captures transactional payment details.
 - **order_id:** Foreign key referencing the order.
 - **payment_type:** Method of payment (e.g., credit_card, boleto, voucher, debit_card).
 - **payment_installments:** Number of monthly installment payments selected by the customer.
 - **payment_value:** Total amount paid.
6. **stg_reviews:** Stores post-purchase customer feedback.
 - **review_id:** Unique identifier of the review.
 - **order_id:** Foreign key referencing the reviewed order.
 - **review_score:** Satisfaction rating ranging from 1 (lowest) to 5 (highest).
 - **review_answer_timestamp:** Timestamp of when the review was submitted.
7. **stg_category_translation:** Reference table translating Portuguese categories to English.
 - **product_category_name:** Category name in Portuguese.
 - **product_category_name_english:** Corresponding English category name.

2.4. Data Quality Challenges & Preprocessing Requirements

The raw dataset contains several quality issues and structural anomalies that must be addressed during the ETL phase to prevent inaccurate business reporting:

- **Missing and Portuguese Product Categories:** A significant number of products have null or empty category names, and the existing ones are in Portuguese. These need to be resolved and mapped to English for professional business reports.
- **Unstructured Timestamps:** Timestamps are formatted as text strings (e.g., DD/MM/YYYY HH:MM). They must be parsed and converted into proper DATETIME formats to enable time-series calculations.
- **Operational Order Statuses:** For core sales analysis, only completed transactions should be processed. Orders that are canceled (canceled), unavailable (unavailable), or in-progress must be filtered out.
- **Payment and Review Duplicates:** A single order can have multiple payment methods or split reviews. The ETL pipeline must implement deduplication strategies (e.g., selecting the primary payment type based on value and the most recent review score) to maintain a clean grain in the Fact table.

PART III: DATA WAREHOUSE DESIGN

3.1. Analytical Model Selection: Star Schema

For this project, a Star Schema has been selected as the design architecture for the Data Warehouse. In dimensional modeling, a Star Schema features a centralized fact table containing quantitative measures, surrounded by multiple denormalized dimension tables describing the business context.

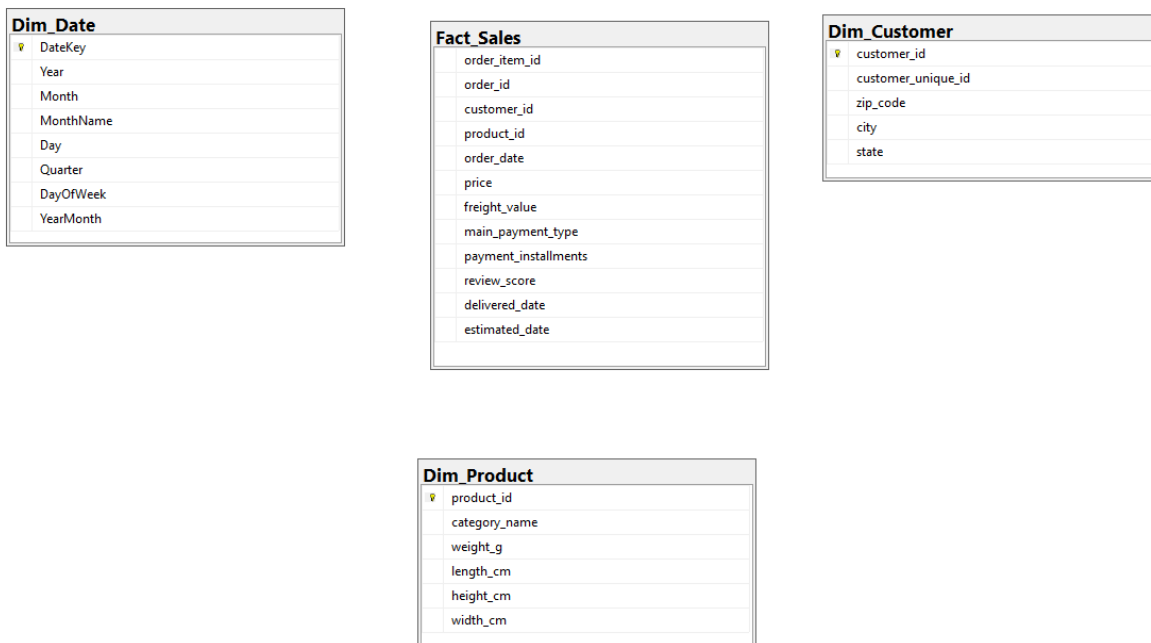
The choice of a Star Schema over a Snowflake Schema is justified by several analytical and performance benefits:

1. **Simplified Queries:** Denormalized dimension tables eliminate the need for complex, multi-level joins (such as joining products to categories, or zip codes to states). This makes SQL queries written for reports simpler and less prone to errors.
2. **Optimized BI Tool Integration:** Standard Business Intelligence tools like Power BI and Tableau are native environments for Star Schemas. They perform optimally when filtering and slicing numerical metrics across direct single-level relationships.

3. **Query Performance:** By minimizing table joins, database execution engines can run aggregations (such as summing sales or averaging delivery times) much faster, which is critical for large datasets.

3.2. Data Warehouse Schema Diagram (ERD)

The Data Warehouse schema connects a central fact table (**Fact_Sales**) to three dimension tables: **Dim_Customer**, **Dim_Product**, and **Dim_Date**.



Star Schema Entity-Relationship Diagram of the Data Warehouse

3.3. Table Architecture and Data Dictionaries

The primary grain of the Data Warehouse is established at the order item level (i.e., each row in the fact table represents a specific item purchased within an order). This allows for highly granular analysis of pricing, shipping, and product-specific characteristics.

Below are the data dictionaries detailing the column structures, data types, keys, and descriptions for each table.

1. Central Fact Table: **Fact_Sales**

This table stores the quantitative business metrics of the sales transactions and connects to all dimensions.

Column Name	Data Type	Key Type	Description
order_id	VARCHAR(50)	Composite PK	Unique ID of the order transaction.
order_item_id	INT	Composite PK	Sequential identifier for items within the same order.
customer_id	VARCHAR(50)	FK	Links to the customer who placed the order.
product_id	VARCHAR(50)	FK	Links to the specific product purchased.
order_date	DATE	FK	Links to Dim_Date (representing purchase date).
price	FLOAT	Measure	Selling price of the item.
freight_value	FLOAT	Measure	Shipping cost charged for this specific item.
main_payment_type	VARCHAR(50)	Attribute	The primary payment method used for the order.
payment_installments	INT	Measure / Attr	Number of monthly installments chosen.

review_score	INT	Measure / Attr	Customer satisfaction rating (1 to 5 stars).
delivered_date	DATETIME2	Attribute	Actual date and time when the customer received the order.
estimated_date	DATETIME2	Attribute	Estimated date and time of delivery promised to the customer.

2. Customer Dimension Table: Dim_Customer

Contains denormalized geographical attributes for analyzing customer distributions.

Column Name	Data Type	Key Type	Description
customer_id	VARCHAR(50)	PK	Unique identifier for the customer record.
customer_unique_id	VARCHAR(50)	Attribute	Persistent identifier to track repeat customer purchases.
zip_code	VARCHAR(15)	Attribute	Zip code prefix of the customer's residence.
city	VARCHAR(100)	Attribute	City name of the customer.
state	VARCHAR(10)	Attribute	State abbreviation (e.g., SP, RJ, MG) in Brazil.

3. Product Dimension Table: Dim_Product

Contains physical specifications and categorization details for products.

Column Name	Data Type	Key Type	Description
product_id	VARCHAR (50)	PK	Unique identifier for the product.
category_name	VARCHAR (100)	Attribute	Translated English name of the product category (e.g., health_beauty).
weight_g	FLOAT	Attribute	Weight of the product in grams.
length_cm	FLOAT	Attribute	Length of the product in centimeters.
height_cm	FLOAT	Attribute	Height of the product in centimeters.
width_cm	FLOAT	Attribute	Width of the product in centimeters.

4. Date Dimension Table: Dim_Date

A dedicated table to support time-intelligence calculations, filtering, and trend lines.

Column Name	Data Type	Key Type	Description
DateKey	DATE	PK	The primary date key representing a unique day.
Year	INT	Attribute	Calendar year (e.g., 2017, 2018).
Month	INT	Attribute	Month number (1 to 12).

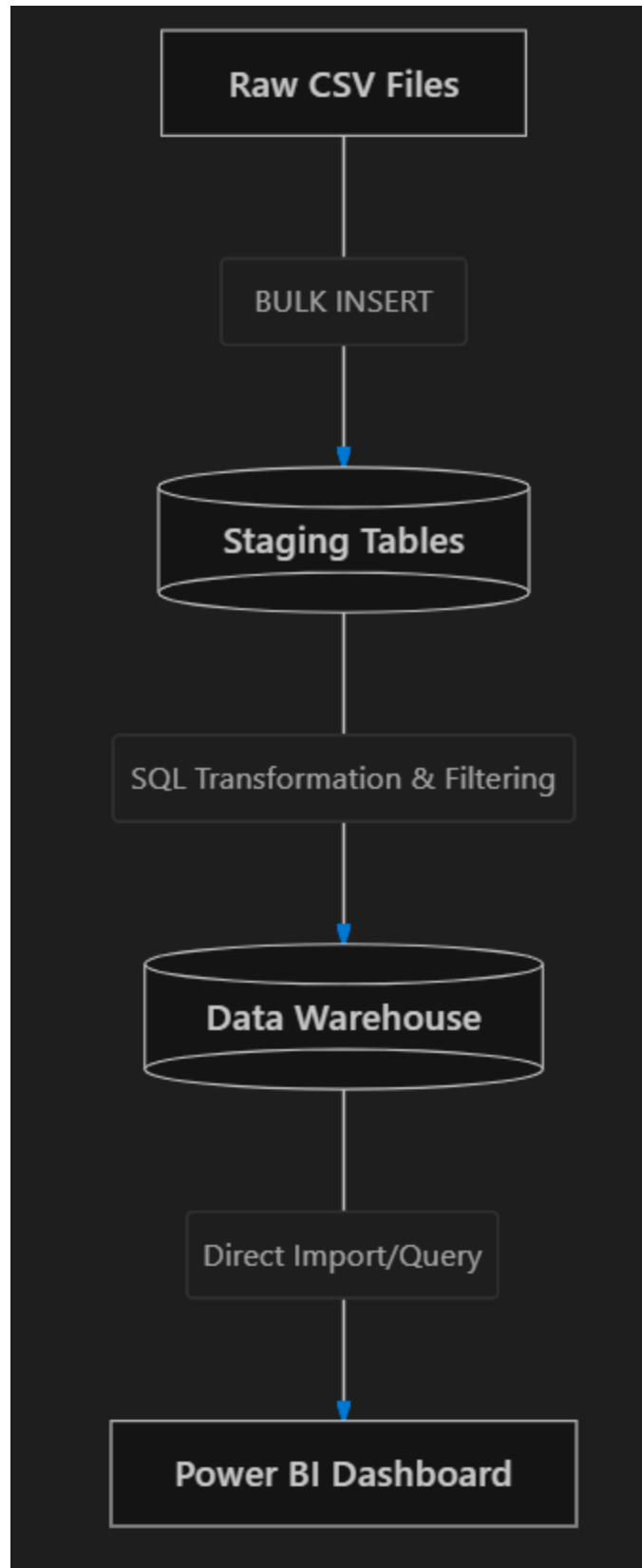
MonthName	VARCHAR(20)	Attribute	Month name in English (e.g., January, February).
Day	INT	Attribute	Day of the month (1 to 31).
Quarter	INT	Attribute	Financial quarter (1 to 4).
DayOfWeek	VARCHAR(20)	Attribute	Name of the weekday (e.g., Monday, Tuesday).
YearMonth	VARCHAR(7)	Attribute	Formatted string for timeline charts (e.g., 2018-07).

PART IV: ETL PROCESS (EXTRACT, TRANSFORM, LOAD)

4.1. ETL Architecture and Pipeline Flow

To build a reliable Data Warehouse, an ETL (Extract, Transform, Load) pipeline was implemented. The data pipeline architecture is structured into three distinct zones:

1. **Raw Files (Source Zone):** Flat CSV files representing raw tables from the Olist dataset located in the olist_csv directory.
2. **Staging Area (Buffer Zone):** Staging tables prefixed with stg_ in the SQL Server database. These tables hold exact copies of the raw CSV files to isolate processing from the source files and avoid parsing issues during complex joins.
3. **Data Warehouse (Presentation Zone):** Cleaned, transformed, and indexed Fact and Dimension tables ready for analytical queries and reporting.



Data Pipeline Diagram

4.2. Extract Phase (E): Raw to Staging

The extraction phase copies data from the local directory into the staging schema of the SQL Server database. Due to the size of the datasets (up to 60MB), SQL Server's high-performance BULK INSERT utility was utilized in the `import_csv_into_staging.sql` script.

An example extraction block for customer data is shown below:

```
BULK INSERT stg_customers
FROM
'D:\College\BI\Project\Business_Intelligence\olist_csv\olist_customers_dataset.csv'
WITH (
    FORMAT = 'CSV',           -- Parse standardized comma-separated values
    FIRSTROW = 2,             -- Skip the header row
    FIELDTERMINATOR = ',',    -- Standard column delimiter
    ROWTERMINATOR = '0x0a',   -- Unix-style line break (LF)
    CODEPAGE = '65001',       -- Force UTF-8 encoding to preserve Brazilian characters
    TABLOCK                   -- Acquire table lock to accelerate bulk insert speed
);
```

4.3. Transform Phase (T): Staging to Star Schema

The transformation logic, written in **ETL_BI.sql**, applies data cleansing, reference translation, type conversions, and deduplication.

1. Time-Intelligence Extraction (Dim_Date)

Timestamps in the raw dataset are imported as strings. During transformation, the **TRY_CONVERT(DATE, NULLIF(..., ''), 103)** function is applied (using code 103 for British/French format DD/MM/YYYY) to parse date elements.

```
SELECT DISTINCT
    TRY_CONVERT(DATE, NULLIF(order_purchase_timestamp, ''), 103) AS [DateKey],
    YEAR(TRY_CONVERT(DATE, NULLIF(order_purchase_timestamp, ''), 103)) AS [Year],
    MONTH(TRY_CONVERT(DATE, NULLIF(order_purchase_timestamp, ''), 103)) AS [Month],
```

```

    DATENAME(MONTH, TRY_CONVERT(DATE, NULLIF(order_purchase_timestamp, ''), 103))
AS [MonthName],
    DAY(TRY_CONVERT(DATE, NULLIF(order_purchase_timestamp, ''), 103)) AS [Day],
    DATEPART(QUARTER, TRY_CONVERT(DATE, NULLIF(order_purchase_timestamp, ''),
103)) AS [Quarter],
    DATENAME(WEEKDAY, TRY_CONVERT(DATE, NULLIF(order_purchase_timestamp, ''),
103)) AS [DayOfWeek],
    FORMAT(TRY_CONVERT(DATE, NULLIF(order_purchase_timestamp, ''), 103),
'yyyy-MM') AS [YearMonth]
INTO Dim_Date
FROM stg_orders
WHERE order_purchase_timestamp IS NOT NULL AND order_purchase_timestamp != ''
AND TRY_CONVERT(DATE, NULLIF(order_purchase_timestamp, ''), 103) IS NOT NULL;

```

2. English Category Translation (Dim_Product)

The native product categories are in Portuguese. To make the BI reports accessible, the staging table **stg_products** is left-joined with **stg_category_translation** to extract English names. A **COALESCE** statement handles missing values by mapping them to 'Unknown'.

```

SELECT
    p.product_id,
    COALESCE(
        NULLIF(t.product_category_name_english, ''),
        NULLIF(p.product_category_name, ''),
        'Unknown'
    ) AS category_name,
    CAST(p.product_weight_g AS FLOAT) AS weight_g,
    CAST(p.product_length_cm AS FLOAT) AS length_cm,
    CAST(p.product_height_cm AS FLOAT) AS height_cm,
    CAST(p.product_width_cm AS FLOAT) AS width_cm
INTO Dim_Product
FROM stg_products p
LEFT JOIN stg_category_translation t
    ON p.product_category_name = t.product_category_name;

```

3. Deduplication and Cardinality Reduction (Fact_Sales)

In e-commerce transactions, orders can have multiple payment methods or multiple post-purchase reviews. This creates a one-to-many relationship that would duplicate transaction values if directly joined. To resolve this, Common Table Expressions (CTEs) utilizing the **ROW_NUMBER()** window function partition the data by `order_id` and rank them:

- For payments: Orders are sorted by **payment_value DESC**, prioritizing the primary payment method (`rn = 1`).
- For reviews: Orders are sorted by **review_answer_timestamp DESC** to retrieve the latest rating score (`rn = 1`).

```
WITH CTE_Payments AS (  
    SELECT  
        order_id,  
        payment_type,  
        TRY_CAST(NULLIF(payment_installments, '') AS INT) AS  
payment_installments,  
        ROW_NUMBER() OVER(PARTITION BY order_id ORDER BY  
TRY_CAST(NULLIF(payment_value, '') AS FLOAT) DESC) as rn  
    FROM stg_payments  
) ,  
CTE_Reviews AS (  
    SELECT  
        order_id,  
        TRY_CAST(NULLIF(review_score, '') AS INT) AS review_score,  
        ROW_NUMBER() OVER(PARTITION BY order_id ORDER BY review_answer_timestamp  
DESC) as rn  
    FROM stg_reviews  
    WHERE review_score IS NOT NULL AND review_score != ''  
)
```

Only orders with status delivered are included to focus on successful sales.

4.4. Load Phase (L): Populating and Constraints

Once the data is transformed, it is written to the destination tables using the **SELECT INTO** command. To establish relational integrity, the scripts alter the tables to define column constraints and bind primary keys:

```
-- Enforce non-nullable constraints on primary key columns
ALTER TABLE Dim_Customer ALTER COLUMN customer_id VARCHAR(50) NOT NULL;
ALTER TABLE Dim_Product ALTER COLUMN product_id VARCHAR(50) NOT NULL;

-- Bind Primary Keys
ALTER TABLE Dim_Customer ADD CONSTRAINT PK_DimCustomer PRIMARY KEY (customer_id);
ALTER TABLE Dim_Product ADD CONSTRAINT PK_DimProduct PRIMARY KEY (product_id);
```

1	SELECT TOP (1000) [customer_id]
2	, [customer_unique_id]
3	, [zip_code]
4	, [city]
5	, [state]
6	FROM [Olist_Ecommerce_DW].[dbo].[Dim_Customer]
7	

90 % No issues found

Results Messages

	customer_id	customer_unique_id	zip_code	city	state
1	00012a2ce6f8dcdca20d059ce98491703	248ffe10d632bebe4f72671f144844c9	6273	osasco	SP
2	000161a058600d5901f007fab4c27140	b0015e09bb4b6e47c52844fab5fb6638	35550	itapecerica	MG
3	0001fd6190edaaf884bcdf3d49edf079	94b11d37cd61cb2994a194d11f89682b	29830	nova venecia	ES
4	0002414f95344307404f0ace7a26f1d5	4893ad4ea28b2c5b3ddf4e82e79db9e6	39664	mendonca	MG
5	000379cdec625522490c315e70c7a9fb	0b83f73b19c2019e182fd552c048a22c	4841	sao paulo	SP
6	0004164d20a9e969af783496f3408652	104bdb7e6a6cdceaa88c3ea5fa6b2b93	13272	valinhos	SP
7	000419c5494106c306a97b5635748086	14843983d4a159080f6afe4b7f346e7c	24220	niteroi	RJ
8	00046a560d407e99b969756e0b10f282	0b5295fc9819d831f68eb0e9a3e13ab7	20540	rio de janeiro	RJ
9	00050bf6e01e69d5c0fd612f1bcfb69c	e3cf594a99e810f58af53ed4820f25e5	98700	ijui	RS
10	000598caf2ef4117407665ac33275130	7e0516b486e92ed3f3afdd6d1276cfbd	35540	oliveira	MG
11	0005aefbb696d34b3424dccc0a0e9fd0	616309b2eeb7bd9c05b0fdfbab28e6c6	3052	sao paulo	SP
12	00062b33cb9f6fe976afdcff967ea74d	f90f55ee274a4ae21510b386134b09cd	2306	sao paulo	SP
13	00066ccbe787a588c52bd5ff404590e3	15090f48004f3b0fc18a167ef82af4db	93525	novo hamb...	RS
14	00072d033fe2e59061ae5c3aff1a2be5	b7c13491fd2aec9308c75bb4de979d8	45026	vitoria da c...	BA
15	0009a69b72033b2d0ec8c69fc70ef768	fa30145b07cad8e972bea351a65216f4	13106	campinas	SP
16	000bf8121c3412d3057d32371c5d3395	1bc9b2dad6aefbfb0c11508e34c8adfc	12335	jacarei	SP
17	000e943451fc2788ca6ac98a682f2f49	d73c3cf4a0922ece1176bd40eb76ac50	99460	colorado	RS
18	000f17e290c26b28549908a04cfe36c1	74541fbb7526dabecd0fc96b1192b9a7	98400	frederico w...	RS
19	000fd45d6fedae68fc6676036610f879	cee6fa72fb403ef9554b41581108084b	12970	piracaia	SP

Dim_Customer

SQL Query Editor (SQL Server) - SQL Query Editor - DW (SQL Server)

```

1 SELECT TOP (1000) [DateKey]
2     , [Year]
3     , [Month]
4     , [MonthName]
5     , [Day]
6     , [Quarter]
7     , [DayOfWeek]
8     , [YearMonth]
9 FROM [Olist_Ecommerce_DW].[dbo].[Dim_Date]
10

```

90 % No issues found

Results Messages

	DateKey	Year	Month	MonthName	Day	Quarter	DayOfWeek	YearMonth
1	2016-09-04	2016	9	September	4	3	Sunday	2016-09
2	2016-09-05	2016	9	September	5	3	Monday	2016-09
3	2016-09-13	2016	9	September	13	3	Tuesday	2016-09
4	2016-09-15	2016	9	September	15	3	Thursday	2016-09
5	2016-10-02	2016	10	October	2	4	Sunday	2016-10
6	2016-10-03	2016	10	October	3	4	Monday	2016-10
7	2016-10-04	2016	10	October	4	4	Tuesday	2016-10
8	2016-10-05	2016	10	October	5	4	Wednesday	2016-10
9	2016-10-06	2016	10	October	6	4	Thursday	2016-10
10	2016-10-07	2016	10	October	7	4	Friday	2016-10
11	2016-10-08	2016	10	October	8	4	Saturday	2016-10
12	2016-10-09	2016	10	October	9	4	Sunday	2016-10
13	2016-10-10	2016	10	October	10	4	Monday	2016-10
14	2016-10-22	2016	10	October	22	4	Saturday	2016-10
15	2016-12-23	2016	12	December	23	4	Friday	2016-12
16	2017-01-05	2017	1	January	5	1	Thursday	2017-01

Query executed successfully

DESKTOP-VIRT5LRVCLCCSD\PTNL

Dim_Date

SQLQuery3.sql..._DW (sa (55))

```

1      SELECT TOP (1000) [product_id]
2          , [category_name]
3          , [weight_g]
4          , [length_cm]
5          , [height_cm]
6          , [width_cm]
7      FROM [Olist_Ecommerce_DW].[dbo].[Dim_Product]
8

```

90 % No issues found

Results Messages

	product_id	category_name	weight_g	length_cm	height_cm	width_cm
1	00066f42aeeb9f3007548bb9d3f33c38	perfumery	300	20	16	16
2	00088930e925c41fd95ebfe695fd2655	auto	1225	55	10	26
3	0009406fd7479715e4bef61dd91f2462	bed_bath_table	300	45	15	35
4	000b8f95fcb9e0096488278317764d19	housewares	550	19	24	12
5	000d9be29b5207b54e86aa1b1ac54872	watches_gifts	250	22	11	15
6	0011c512eb256aa0dbbb544d8dffcf6e	auto	100	16	15	16
7	00126f27c813603687e6ce486d909d01	cool_stuff	700	25	5	15
8	001795ec6f1b187d37335e1c4704762e	consoles_games	600	30	20	20
9	001b237c0e9bb435f2e54071129237e9	bed_bath_table	6000	40	4	30
10	001b72dfd63e9833e8c02742adf472e3	furniture_decor	600	26	8	22
11	001c5d71ac6ad696d22315953758fa04	bed_bath_table	1800	47	21	41
12	00210e41887c2a8ef9f791ebc780cc36	health_beauty	300	30	15	15
13	002159fe700ed3521f46cfcf6e941c76	fashion_shoes	1850	36	37	16
14	0021a87d4997a48b6cef1665602be0f5	computers_acc...	650	25	10	15
15	00250175f79f584c14ab5cecd80553cd	housewares	750	30	5	35
16	002552c0663708129c0019cc97552d7d	cool_stuff	355	22	16	12
17	002959d7a0b0990fe2d69988affcbc80	furniture_decor	2600	105	3	70
18	002af88741ba70c7b5cf4e4a0ad7ef85	cool_stuff	600	40	30	40

Query executed successfully. DESKTOP-VJRT5LB\CLCCSDLPTNH... sa (55) Olist

Dim_Product

2

SELECT TOP (1000) [order_item_id]

3

[order_id]

4

[customer_id]

5

[product_id]

6

[order_date]

7

[price]

8

[freight_value]

9

[main_payment_type]

10

[payment_installments]

11

[review_score]

12

[delivered_date]

13

[estimated_date]

14

FROM [Olist_Ecommerce_DW].[dbo].[Fact_Sales]

90 %

No issues found

Ln: 1, Ch: 1

SPC

CRLF

Windows 1252

Results

Messages

	order_item_id	order_id	customer_id	product_id	order_date	price	freight_value	main_payment_type	payment_installments	review_score	delivered
1	1	00054e8431b9d7675808bcb819fb4a32	32e2efab09e778d99bf2e0ecd4898718	8d4f2bb7e93e6710a28f34fa83ee7d28	2017-12-10	19.9	11.85	credit_card	1	4	2017-12-
2	1	0006288aa423d2a3f00fcb17cd7d8719	2355af7c75e7c98b43a87b2a7f210dc5	368c6c730842d78016ad823897a372db	2018-02-13	49.9	13.37	boleto	1	5	2018-02-
3	2	0006288aa423d2a3f00fcb17cd7d8719	2355af7c75e7c98b43a87b2a7f210dc5	368c6c730842d78016ad823897a372db	2018-02-13	49.9	13.37	boleto	1	5	2018-02-
4	1	000a6d2e52bad2f9db70584c5a2ded	fff5169e583fd07fac9fec88962f1189d	4fa33915031a8cde03dd0d3e8fb2701	2018-05-11	144	8.77	credit_card	1	1	2018-05-
5	1	000c3e6f12759851cc3cbb4b83257896	3773bdf1a6fbd29233ea1c1b573c4f22	b50c950aba0dcead2c48032a690ce817	2017-08-12	99	13.71	boleto	1	5	2017-08-
6	1	001021efa8636c29475e7734483457d	2dfbf74859104caf100df3720a1d833d	5d7c23067ed3fc8c6e699b9373d5890b	2018-02-27	49	15.1	credit_card	3	4	2018-03-
7	1	001021efa8636c29475e7734483457d	2dfbf74859104caf100df3720a1d833d	5d7c23067ed3fc8c6e699b9373d5890b	2018-02-27	49	15.1	credit_card	3	4	2018-03-
8	1	001e7ba991be1b16605ca0316e7130f9	b6a1445ee5053a05c827e7a5188ad02	884fa3cd42986ba480ea2f8ae4e25ff7	2017-03-18	195	18.21	credit_card	10	5	2017-03-
9	1	0020262c8a370bd5a174a6a2a267321	9ec353f970bdf785f6568724d9ea19aa	a5341e3f8155dbb3e6d323d3ea289729	2017-11-28	79.5	21.05	credit_card	2	4	2017-12-
10	1	0020a2225eb79a372d0fe0e3cca688	0c45155afdbff9622c40824057f9b34	e19ddcc85537b41f22116c8d5425ef46	2017-08-15	29.99	15.1	credit_card	4	5	2017-08-
11	1	0025081dcf9309a5052ae82c6e396	7e6fefa11d6a2c832593264f69df9de3	4e3f399366b0047a572b6682f9bb166e	2018-03-27	14.95	18.23	credit_card	3	5	2018-04-

SQLQuery4.s...DW (sa (66))

SQLQuery3.sq...DW (sa (55))

SQLQuery2.sq...DW (sa (53))

SQLQuery1.sq...DW (sa (70))

1

SELECT TOP (1000) [order_item_id]

2

[order_id]

3

[customer_id]

4

[product_id]

5

[order_date]

6

[price]

7

[freight_value]

8

[main_payment_type]

9

[payment_installments]

10

[review_score]

11

[delivered_date]

12

[estimated_date]

13

FROM [Olist_Ecommerce_DW].[dbo].[Fact_Sales]

14

90 %

No issues found

Ln: 1, Ch: 1

SPC

CRLF

Windows 1252

Results

Messages

	customer_id	product_id	order_date	price	freight_value	main_payment_type	payment_installments	review_score	delivered_date	estimated_date
1	32	32e2efab09e778d99bf2e0ecd4898718	8d4f2bb7e93e6710a28f34fa83ee7d28	2017-12-10	19.9	11.85	credit_card	1	4	2017-12-18 22:03:00.0000000
2	9	2355af7c75e7c98b43a87b2a7f210dc5	368c6c730842d78016ad823897a372db	2018-02-13	49.9	13.37	boleto	1	5	2018-02-26 13:55:00.0000000
3	9	2355af7c75e7c98b43a87b2a7f210dc5	368c6c730842d78016ad823897a372db	2018-02-13	49.9	13.37	boleto	1	5	2018-02-26 13:55:00.0000000
4	ad	fff5169e583fd07fac9fec88962f1189d	4fa33915031a8cde03dd0d3e8fb2701	2018-05-11	144	8.77	credit_card	1	1	2018-05-18 16:46:00.0000000
5	86	3773bdf1a6fbd29233ea1c1b573c4f22	b50c950aba0dcead2c48032a690ce817	2017-08-12	99	13.71	boleto	1	5	2017-08-19 15:22:00.0000000
6	7d	2dfbf74859104caf100df3720a1d833d	5d7c23067ed3fc8c6e699b9373d5890b	2018-02-27	49	15.1	credit_card	3	4	2018-03-08 20:06:00.0000000
7	57	2013d892495e1a101d742d533d2d1119	c389f712c4b4510bc9970ee93e8b1a28	2018-01-16	289	26.33	credit_card	2	4	2018-01-26 22:14:00.0000000
8	99	b6a1445ee5053a05c827e7a5188ad02	884fa3cd42986ba480ea2f8ae4e25ff7	2017-03-18	195	18.21	credit_card	10	5	2017-03-28 14:23:00.0000000
9	21	9ec353f970bdf785f6568724d9ea19aa	a5341e3f8155dbb3e6d323d3ea289729	2017-11-28	79.5	21.05	credit_card	2	4	2017-12-02 12:27:00.0000000
10	8	0c45155afdbff9622c40824057f9b34	e19ddcc85537b41f22116c8d5425ef46	2017-08-15	29.99	15.1	credit_card	4	5	2017-08-24 19:19:00.0000000
11	6	7e6fefa11d6a2c832593264f69df9de3	4e3f399366b0047a572b6682f9bb166e	2018-03-27	14.95	18.23	credit_card	3	5	2018-04-05 23:48:00.0000000

Fact_Sales

PART V: DASHBOARD IMPLEMENTATION (BI DASHBOARD)

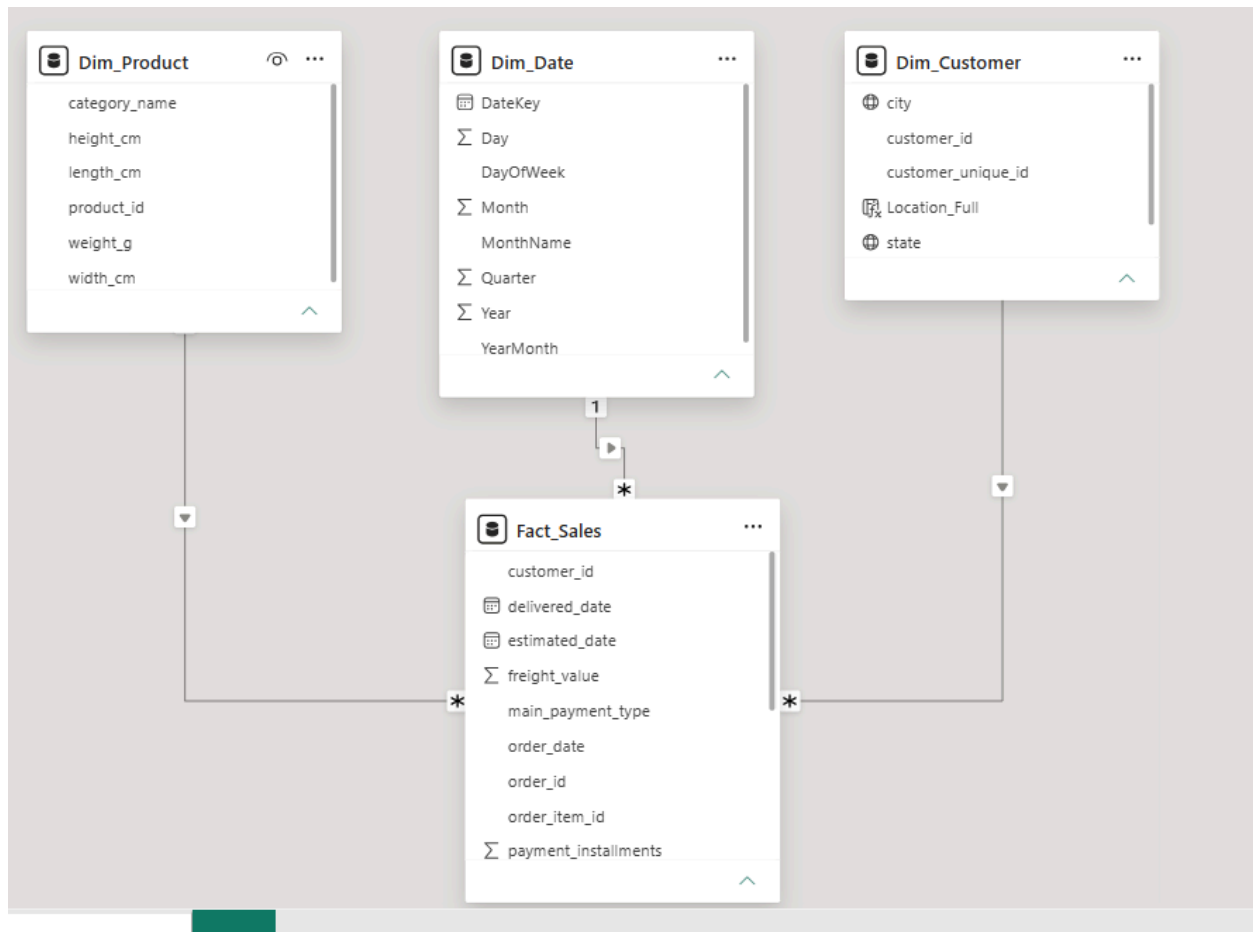
5.1. Tool Selection and Data Connection

For the visualization layer, **Power BI Desktop** was selected as the Business Intelligence tool. **Power BI** excels at handling dimensional models, supporting advanced DAX calculations, and providing highly interactive, cross-filtered reporting interfaces.

The connection to the **SQL Server Data Warehouse** was established using **Import Mode** to load tables into **Power BI's In-Memory xVelocity** database engine. This mode ensures maximum query performance and unlocks the full suite of **DAX** time-intelligence functions.

Once imported, the active relationships matching the Star Schema were confirmed:

- Fact_Sales[customer_id] → Dim_Customer[customer_id] (Many-to-One, Single Filter Direction)
- Fact_Sales[product_id] → Dim_Product[product_id] (Many-to-One, Single Filter Direction)
- Fact_Sales[order_date] → Dim_Date[DateKey] (Many-to-One, Single Filter Direction)



Power BI Data Model relationship diagram

5.2. DAX Measure Development

To support business analysis, **key performance indicators (KPIs)** and operational metrics were developed using **Data Analysis Expressions (DAX)**. Below are the primary measures created:

1. **Total Sales Revenue:** Calculates the gross product sales value.

$$\text{Total Revenue} = \text{SUM}(\text{Fact_Sales}[\text{price}])$$

2. **Total Orders:** Computes the distinct count of unique orders processed.

$$\text{Total Orders} = \text{DISTINCTCOUNT}(\text{Fact_Sales}[\text{order_id}])$$

3. **Total Customers:** Counts the total number of unique customers using their unique identity key.

$$\text{Total Customers} = \text{DISTINCTCOUNT}(\text{Dim_Customer}[\text{customer_unique_id}])$$

4. **Average Satisfaction Score:** Calculates the mean customer review score.

$$\text{Avg Satisfaction} = \text{AVERAGE}(\text{Fact_Sales}[\text{review_score}])$$

5. **Average Delivery Days:** Measures the average shipping duration in days from purchase to delivery.

$$\text{Avg Delivery Days} = \text{AVERAGE}(\text{DATEDIFF}(\text{Fact_Sales}[\text{order_date}], \text{Fact_Sales}[\text{delivered_date}], \text{DAY}))$$

6. **Freight-to-Price Ratio:** Measures the percentage of shipping costs relative to product price to evaluate logistics overhead.

$$\% \text{ Freight to Price} = \text{DIVIDE}(\text{SUM}(\text{Fact_Sales}[\text{freight_value}]), \text{SUM}(\text{Fact_Sales}[\text{price}]), 0)$$

5.3. Dashboard Pages and Visual Layouts

Each cluster groups related visualizations together to answer specific business questions, allowing users to analyze sales, products, logistics, and controls independently.

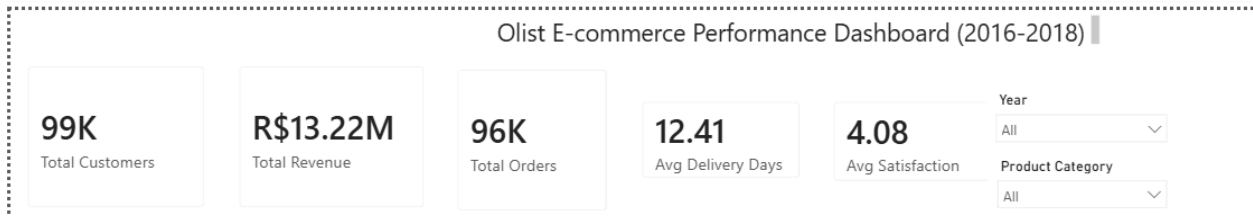
1. Control & Filter Panel Dashboard

This section acts as the control center of the dashboard, allowing users to slice and query the data interactively.

- **Visual Components:**

- **Report Title Textbox:** A static text header at the top left of the canvas displaying the report name ("Olist E-Commerce Analytics Dashboard").
- **Year Slicer:** A dropdown selection panel bound to **Dim_Date[Year]**, which filters the entire report's data for a selected calendar year (2016, 2017, or 2018).

- **Product Category Slicer:** A dropdown selection panel bound to **Dim_Product[category_name]**, enabling users to filter all charts by specific product categories.
- **Business Purpose:** To provide dynamic scoping and slice-and-dice capability across the transaction timeframe and product categories.

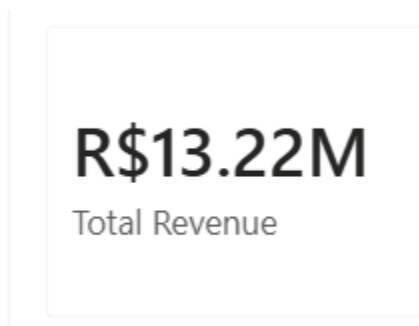


Control & Filter Panel Dashboard

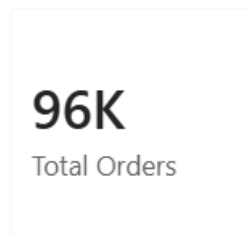
2. Sales & Customer Overview Dashboard

This cluster focuses on core financial performance, customer acquisition volume, time-series trends, and transactional payment methods.

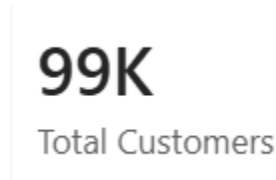
- **Visual Components:**
 - **Total Revenue Card:** Displays the overall sales value (Total Revenue) as a primary financial KPI.



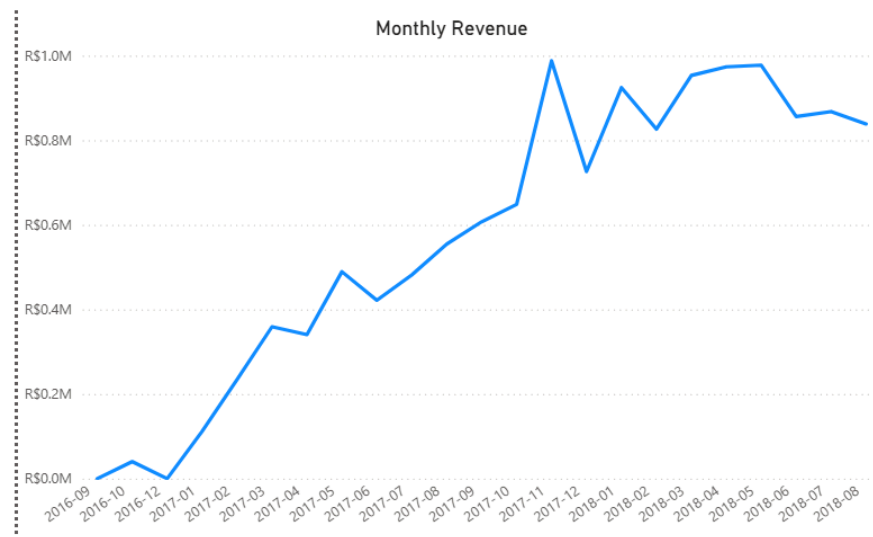
- **Total Orders Card:** Displays the total volume of successful transactions (Total Orders) to track transaction throughput.



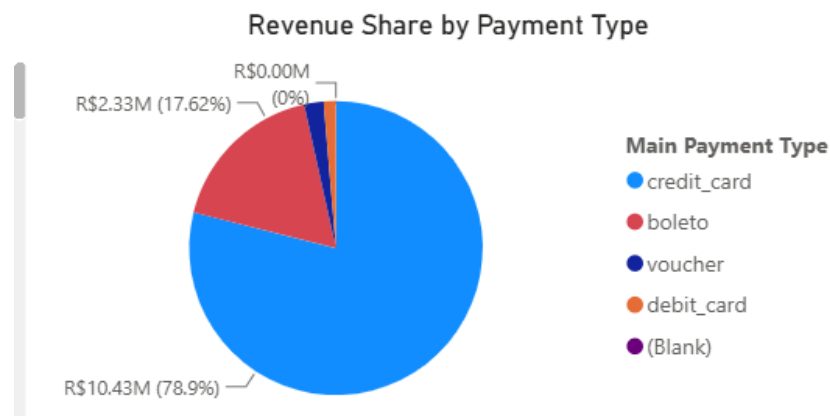
- **Total Customers Card:** Displays the number of unique buyers (Total Customers) to evaluate customer acquisition.



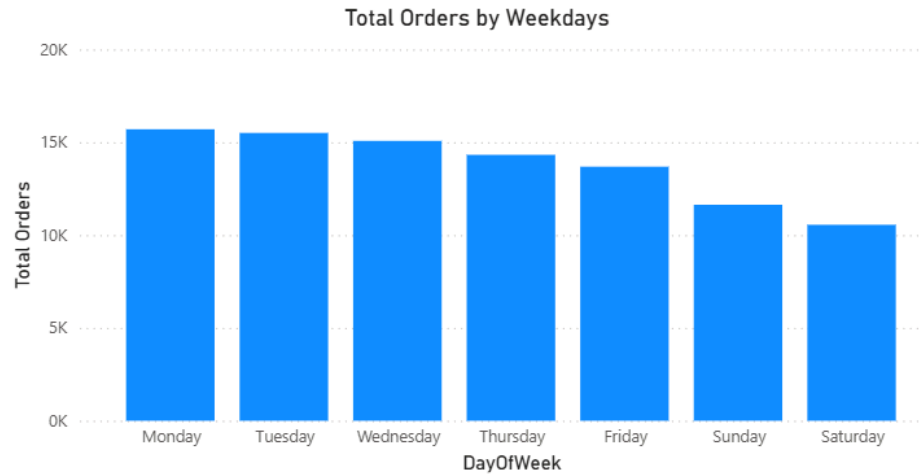
- **Sales Revenue Trend (Line Chart):** Plots Total Revenue (Y-axis) against YearMonth (X-axis) to show sales growth over time and monthly fluctuations.



- **Revenue by Payment Method (Pie Chart):** Represents the percentage share of Total Revenue generated by each payment method (**main_payment_type** - e.g., credit card, boleto, voucher, debit card).



- **Order Count by Weekday (Column Chart):** Plots Total Orders across the days of the week (DayOfWeek) to identify the most active shopping days.

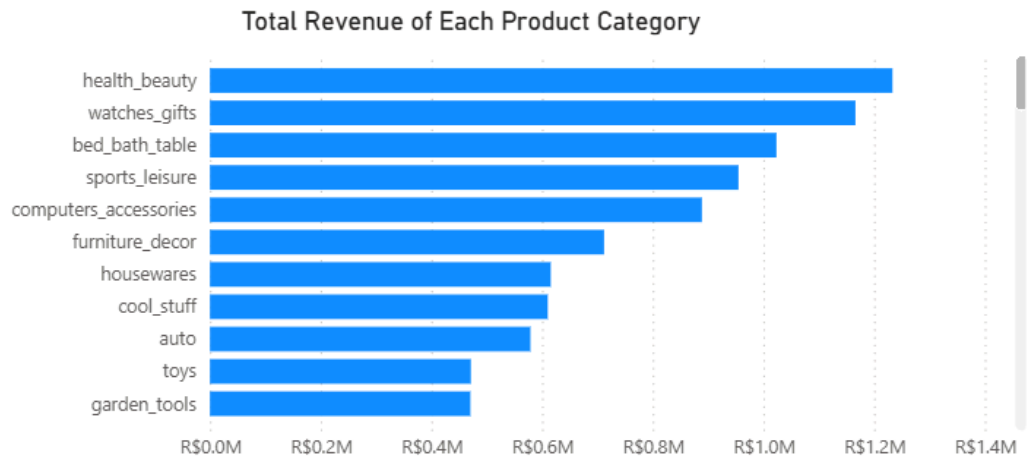


- **Business Purpose:** To monitor sales and customer acquisition trends, evaluate payment preferences, and analyze hourly/daily shopping behavior to plan marketing campaigns.

3. Product Performance & Consumer Behavior Dashboard

This cluster connects product categories with checkout installment preferences and customer feedback.

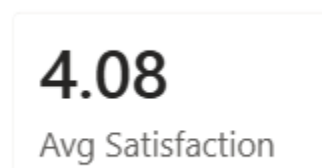
- **Visual Components:**
 - **Top Product Categories (Bar Chart):** A horizontal bar chart displaying the top product categories (**category_name**) ranked by Total Revenue.



- **Orders by Installments (Column Chart):** A vertical bar chart showing Total Orders grouped by the number of installment payments (`payment_installments`).



- **Customer Satisfaction KPI Card:** A card visual displaying the overall satisfaction rating (Avg Satisfaction).



- **Business Purpose:** To identify high-revenue product lines, evaluate customer preference for installment financing, and monitor average satisfaction levels to detect quality issues.

4. Logistics & Regional Operations Dashboard

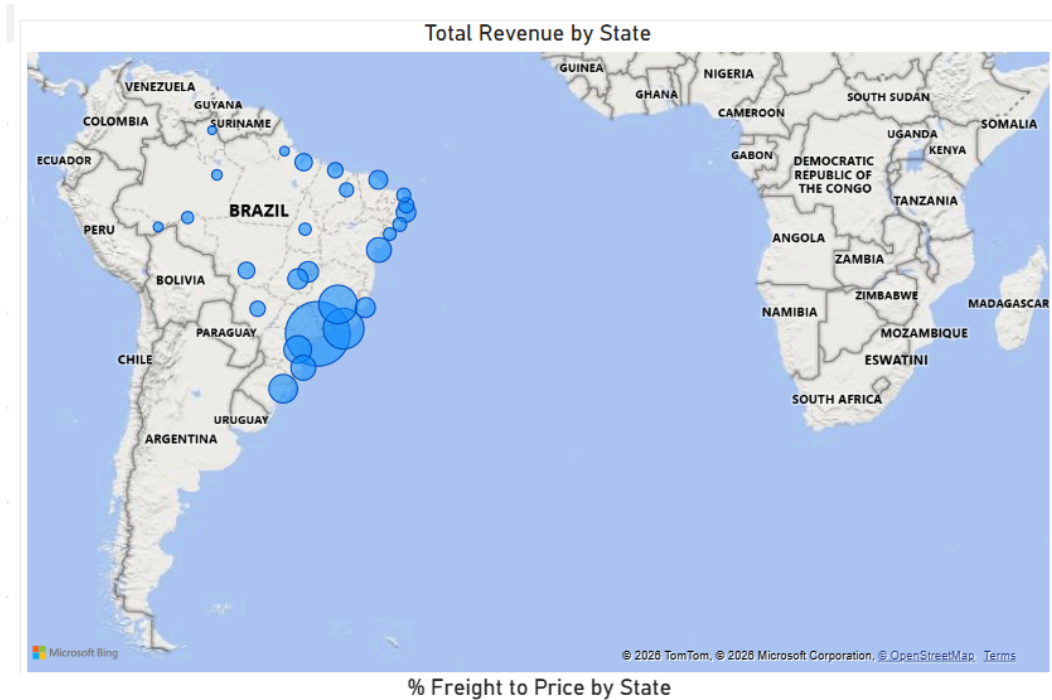
This cluster focuses on shipping speed, regional delivery footprints, and logistics cost overheads.

- **Visual Components:**

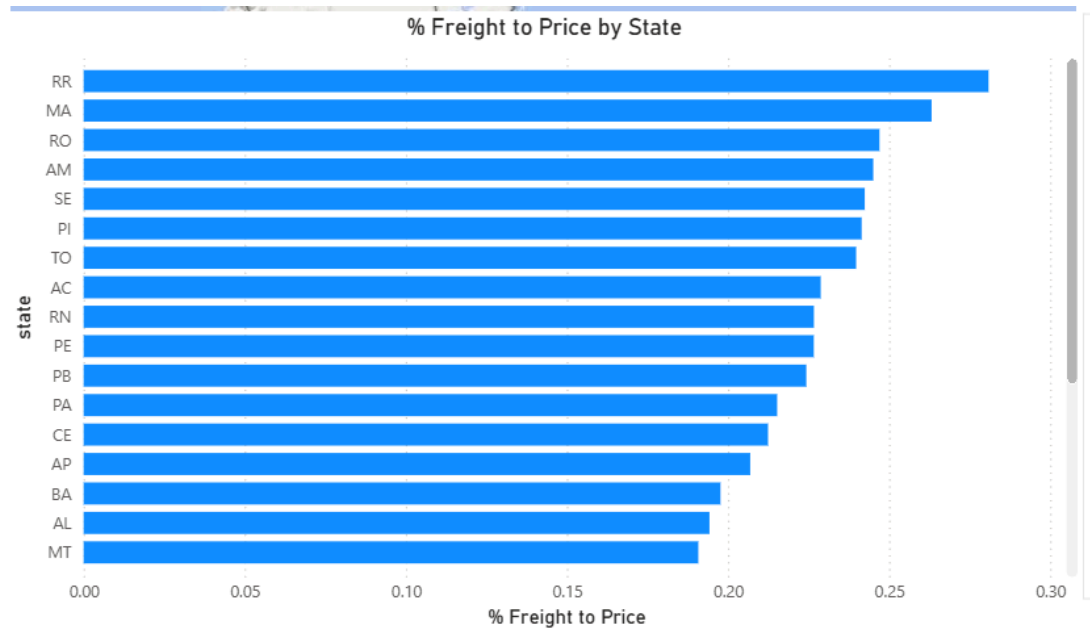
- **Average Delivery Days KPI Card:** A card visual displaying the average time taken from order to delivery (Avg Delivery Days).



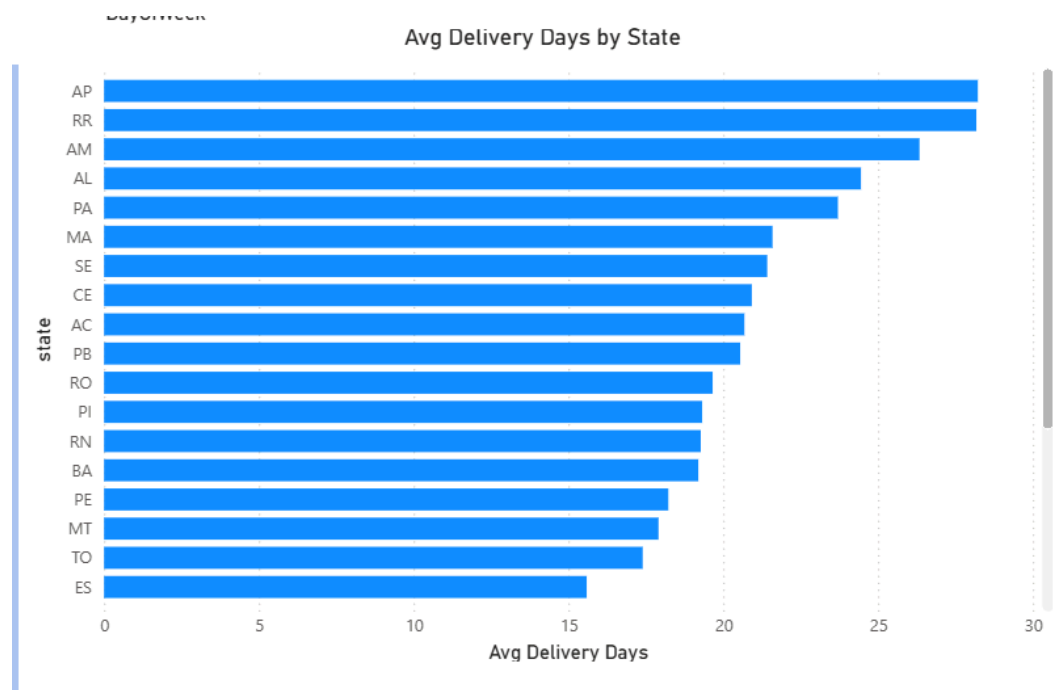
- **Geographical Sales Distribution (Map):** A map plotting customer cities (Location_Full), where bubble sizes represent the volume of Total Revenue.



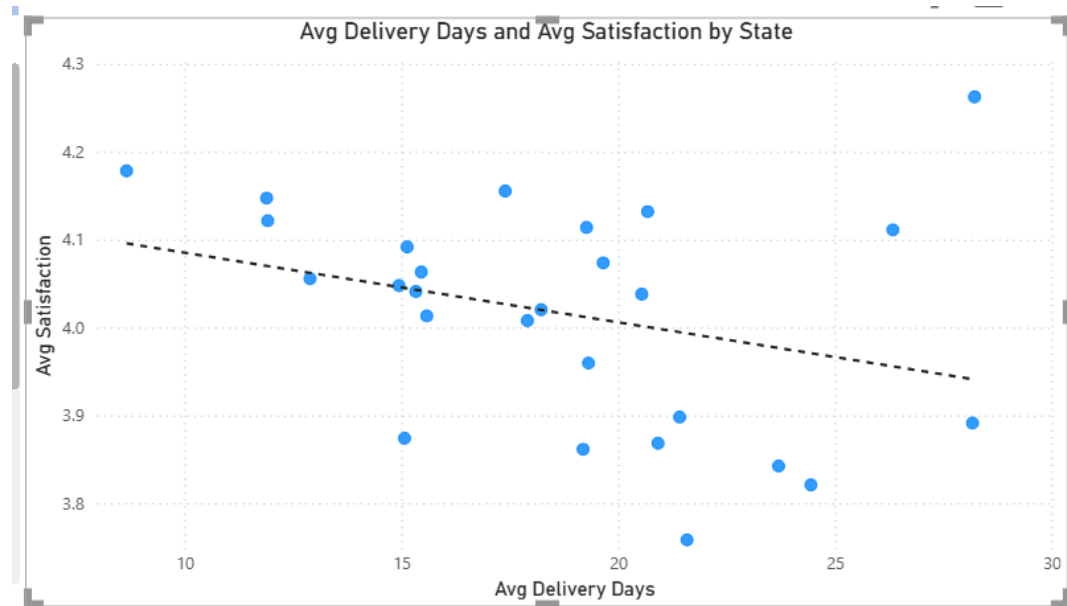
- **Regional Logistics Overhead (Bar Chart):** A horizontal bar chart plotting the ratio of shipping cost to product price (% Freight to Price) by customer state (state).



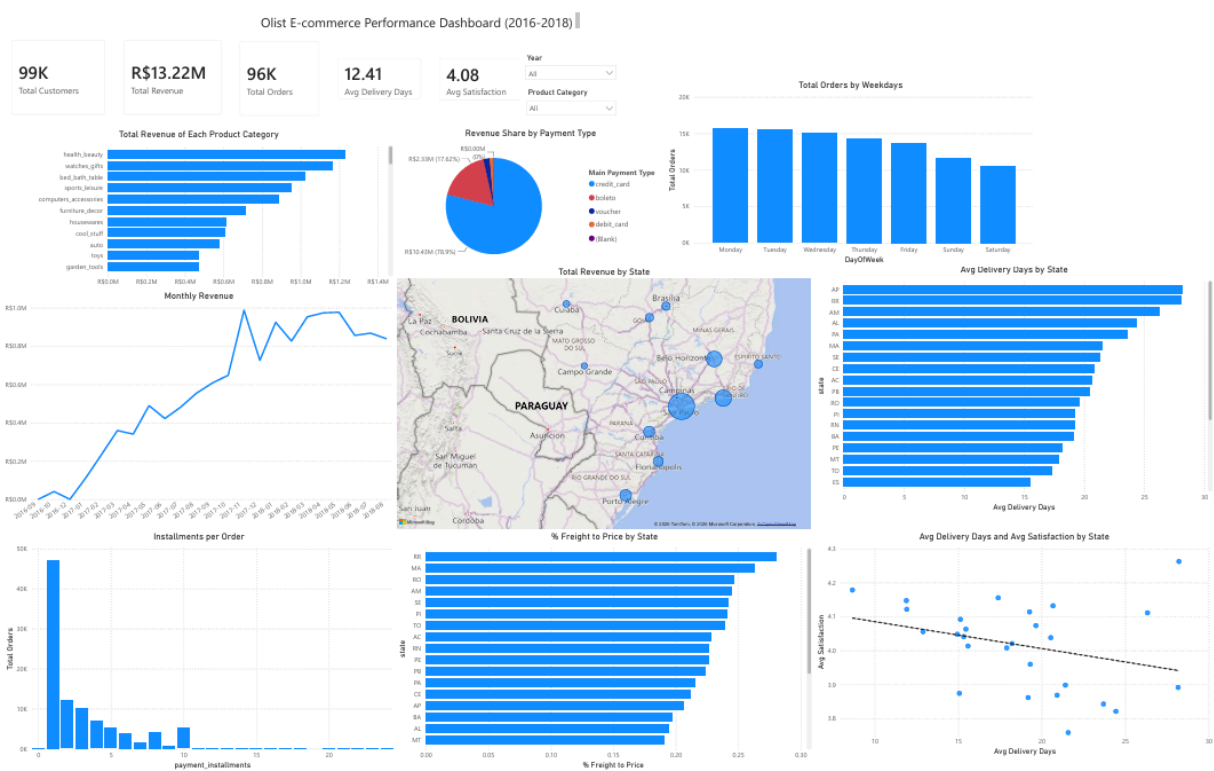
- **Average Delivery Days by State (Bar Chart):** A horizontal bar chart displaying the mean delivery duration (Avg Delivery Days) across different states (state).



- **Satisfaction vs. Delivery Duration Correlation (Scatter Chart):** Plots Avg Satisfaction against Avg Delivery Days grouped by state (state). This visual highlights the geographical areas where delivery speed directly degrades or enhances customer ratings.



- **Business Purpose:** To pinpoint states with excessive shipping fees or shipping delays, evaluate delivery carrier speed per state, and prove the direct correlation between delivery performance and customer ratings.



Full dashboard

PART VI: INSIGHT ANALYSIS

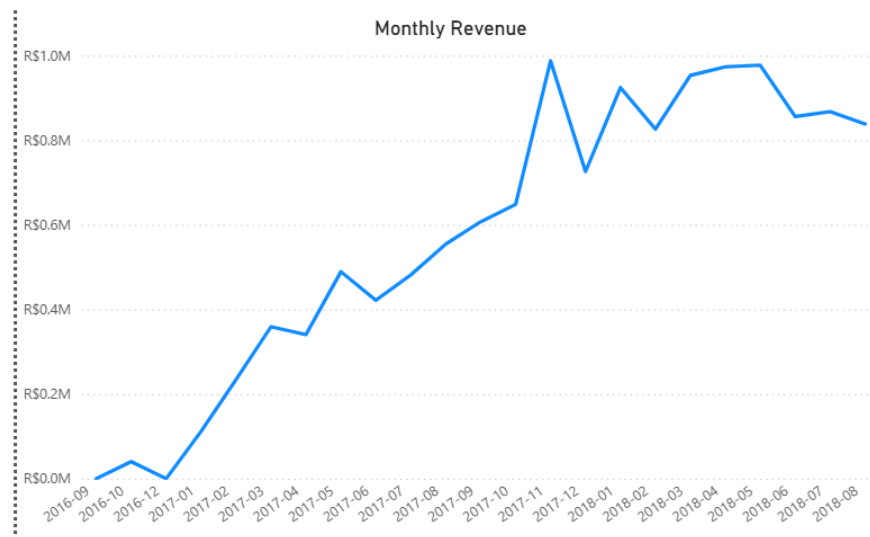
This section presents the key findings and analytical insights derived from the Olist e-commerce dataset, covering 96,478 successfully delivered orders totalling R\$13,221,498 in gross revenue across Brazil from 2016 to 2018. Each insight is supported by concrete figures computed directly from the source data.

6.1. Sales Revenue & Seasonal Trends

Insight 1: Strong Revenue Performance in Early 2018 with a Peak in May. Monthly revenue analysis for 2018 reveals a sustained high level of sales from January through May. The platform reached its highest monthly revenue of **R\$977,545** in May 2018. Interestingly, while January recorded the highest transaction volume (7,069 orders), May generated the most revenue from fewer orders (6,749 orders), indicating a significantly higher Average Order Value (AOV) during this month. Following the May peak, revenue experienced a noticeable drop of approximately 12.4% going into June (R\$856,078) and plateaued around the R\$830k–R\$860k range through August.

Month	Revenue (R\$)	Orders
Jan 2018	924,645	7,069
Feb 2018	826,437	6,555
Mar 2018	953,356	7,003
Apr 2018	973,534	6,798
May 2018	977,545	6,749
Jun 2018	856,078	6,099
Jul 2018	867,953	6,159

Business Recommendation: The revenue surge in May is deeply tied to consumer behavior in Brazil, specifically driven by **Mother's Day (Dia das Mães)**—historically the second biggest retail event of the year in the country after Black Friday. To counteract the mid-year sales dip observed from June to August, Olist should proactively implement targeted winter clearance campaigns or bundle promotions to maintain momentum during these cooler months.

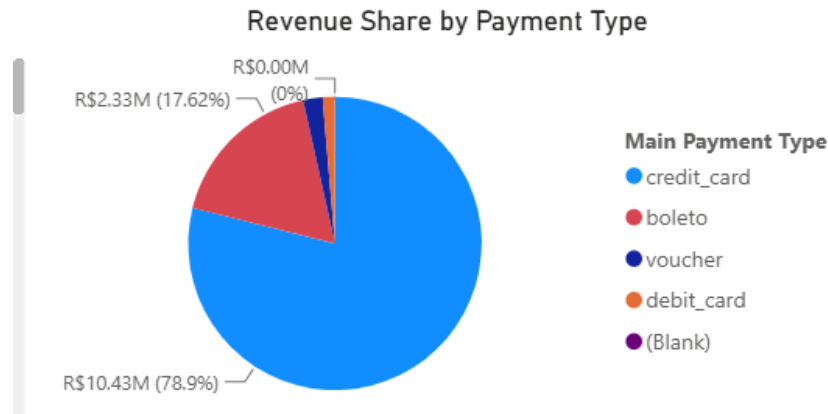


6.2. Consumer Shopping Behavior & Payment Preferences

Insight 2: Credit Cards Dominate, Indicating Consumer Trust in Digital Payments. Of the total **R\$13,221,498** processed across all delivered orders, credit cards account for **R\$10,431,457 (78.90%)** of all transactions—an overwhelmingly dominant payment channel. The second most popular method is *boleto bancário* (a Brazilian cash-based bank slip) at **R\$2,329,808 (17.62%)**. Digital debit cards and vouchers together represent only **3.48%** of the total transaction value.

Payment Method	Revenue (R\$)	Share (%)
Credit Card	10,431,457	78.90%
Boleto	2,329,808	17.62%

Voucher	282,230	2.13%
Debit Card	177,868	1.35%

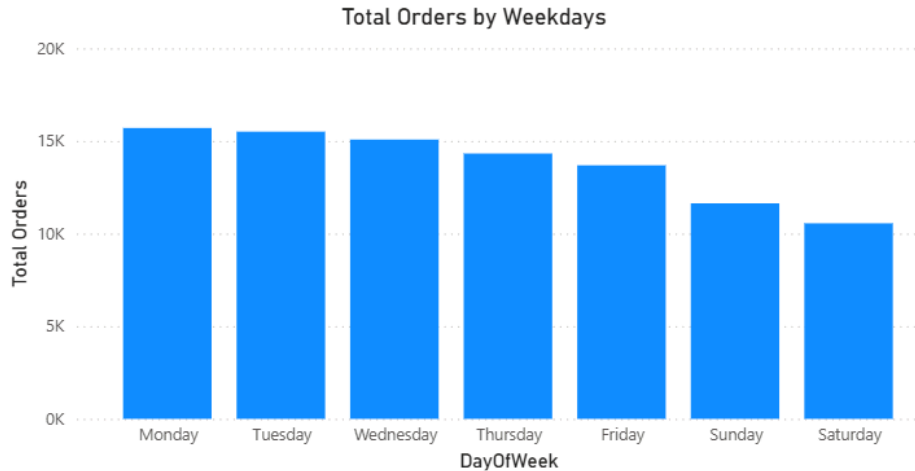


Insight 3: Installment Financing is a Major Driver for Sales Conversion. Among the 96,478 delivered orders, single-payment transactions are the most common, accounting for **47,011 orders (48.73%)**. However, over half of the customer base relies on installment plans to complete their purchases. Specifically, 2 installments (11,928 orders) and 3 installments (10,093 orders) are highly popular. Notably, there is a distinct spike at **10 installments (5,133 orders)**, which significantly exceeds the 6, 7, and 9-month plans. This highlights a strong cultural preference for long-term financing in the Brazilian retail market.



Insight 4: Shopping Activity Peaks at the Beginning of the Week. Analysis of order timing across the 96,478 delivered orders reveals a clear weekly pattern where shopping activity is highly concentrated early in the workweek, contrary to the assumption of weekend shopping.

- **Monday** leads the entire week with the highest volume at **15,701 orders**, closely followed by **Tuesday (15,503)** and **Wednesday (15,076)**.
- As the week progresses, transaction volumes gradually decline, dropping significantly over the weekend to reach their lowest points on **Saturday (10,555 orders)** and **Sunday (11,635 orders)**.
- This trend indicates that Brazilian consumers actively browse and complete e-commerce purchases at the start of their workweek, rather than during their weekend leisure time.



Business Recommendation: To fully capitalize on these consumer patterns, Olist should implement a synchronized marketing and operational strategy:

- **Timing & Budget Allocation:** Front-load the digital marketing budget and schedule major promotional blasts (emails, push notifications, flash sales) for **Sunday evenings and Monday mornings** to capture the early-week shopping peak. Conversely, ad spend should be significantly scaled back on Saturdays to prevent budget waste.
- **Financing as a Marketing Tool:** Since long-term financing is a major conversion driver, marketing creatives and product pages must prominently feature the "**10x sem juros**" (**10 months interest-free**) option, especially for high-ticket items.
- **Payment Infrastructure:** While maintaining a frictionless and secure credit card checkout experience is the top priority, Olist must continue supporting *Boleto* to cater to the 17.6% of the market that still relies on cash-based digital payments.

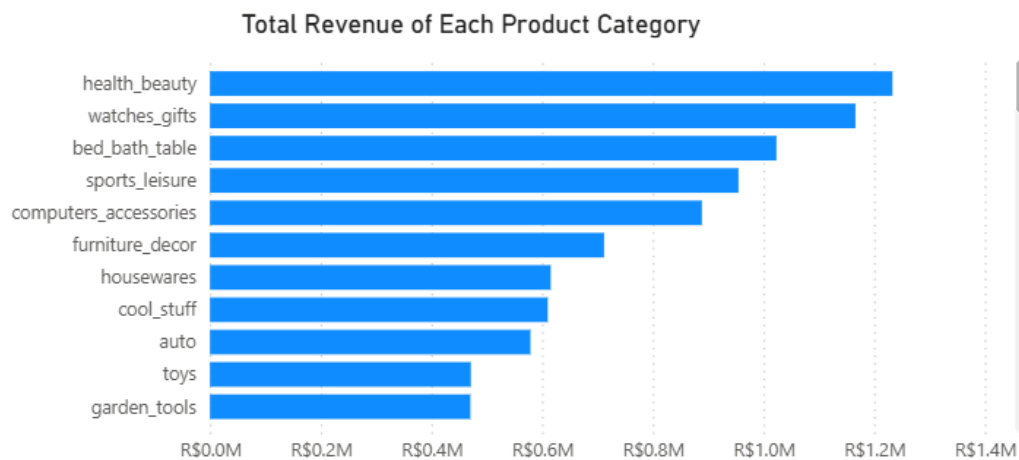
6.3. Product Category Performance

Insight 5: Health & Beauty and Watches/Gifts Lead Revenue — But Not Always Volume. The top 5 product categories by revenue tell an interesting story about unit economics:

Rank	Category	Revenue (R\$)	Orders	Avg Revenue/Order

1	health_beauty	1,233,132	8,647	R\$ 142.6
2	watches_gifts	1,166,177	5,495	R\$ 212.2
3	bed_bath_table	1,023,435	9,272	R\$ 110.4
4	sports_leisure	954,853	7,530	R\$ 126.8
5	computers_accessories	888,725	6,530	R\$ 136.1

watches_gifts generates the second-highest total revenue with notably **fewer orders than health_beauty and bed_bath_table**, implying customers spend significantly more per transaction in this category (R\$212 average vs R\$142 for health_beauty). This makes **watches_gifts a high-value, low-volume category, while bed_bath_table is a high-volume, lower-value category.**



Business Recommendation: Apply differentiated marketing strategies — luxury-oriented promotions for watches_gifts to maintain premium perception, and loyalty/bundle promotions for bed_bath_table to increase volume further.

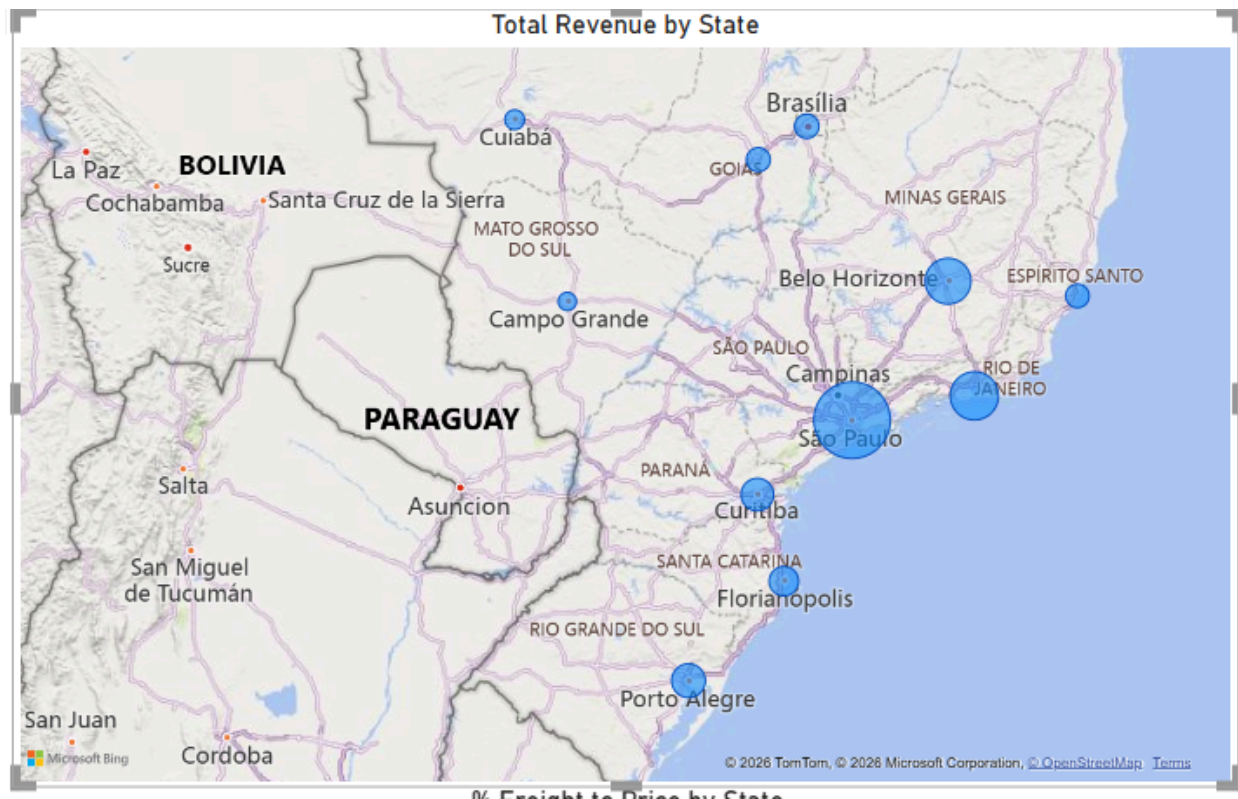
6.4. Geographic Distribution & Regional Market Concentration

Insight 6: São Paulo Dominates the Market with 42% of All Orders. Brazil's geographic revenue distribution is highly concentrated. The state of São Paulo (SP)

accounts for **R\$5,067,633**—representing **38.33%** of total revenue—processed across 40,501 orders (41.98% of all delivered orders). The next two states, Rio de Janeiro (RJ) and Minas Gerais (MG), collectively add another R\$3,312,133. This means the top 3 states generate **over 63%** of all revenue, illustrating a heavy reliance on the southeastern economic hub.

Rank	State	Revenue (R\$)	Orders	Freight/Revenue
1	SP	5,067,633	40,501	13.9%
2	RJ	1,759,651	12,350	16.8%
3	MG	1,552,482	11,354	17.2%
4	RS	728,897	5,345	18.2%
5	PR	666,064	4,923	17.4%

Notably, SP also benefits from the lowest freight-to-revenue ratio (**14.0%**) due to its geographic proximity to major distribution centers. In contrast, states farther south like RS carry a higher logistics burden (**18.0%**), which can erode profit margins and deter price-sensitive customers.



Business Recommendation: To capture growing market demand outside of São Paulo while reducing freight overheads, Olist should evaluate expanding warehouse capacity or partnering with regional fulfillment centers in RJ and MG. Bringing logistics hubs closer to these secondary markets will help drive their freight-to-price ratios below the current 17.0% average.

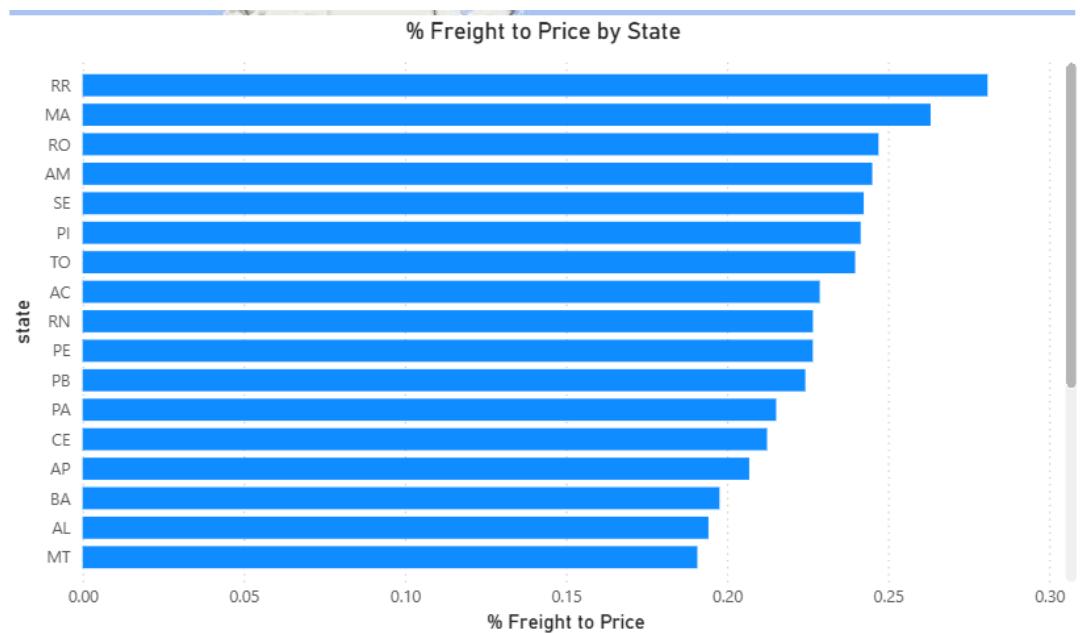
6.5. Logistics Efficiency & Shipping Cost Burden

Insight 7: Remote Northern States Suffer from Disproportionately High Shipping Costs. The freight-to-revenue ratio (the percentage of product price consumed by shipping fees) reveals a significant regional logistics disparity:

State	Freight/Revenue	Revenue (R\$)
RR (Roraima)	28.1%	7,057

MA (Maranhão)	26.3%	117,009
RO (Rondônia)	24.7%	45,683
AM (Amazonas)	24.5%	22,156
SE (Sergipe)	24.2%	56,574

For customers in **Roraima (RR)**, nearly **1 in every 3.6 reais spent on a product goes directly to shipping**. This is more than double the freight burden of customers in São Paulo (13.9%). Such high logistical costs are a major barrier to customer acquisition and retention in these regions.



Insight 8: Average Delivery Time is 12.41 Days, with Severe Regional Disparities. Across all delivered orders, the average order-to-delivery duration is **12.41 days**. However, this overall average conceals a dramatic geographic disparity between the fastest and slowest states, heavily influenced by Brazil's vast territorial challenges.

Fastest States by Average Delivery Speed

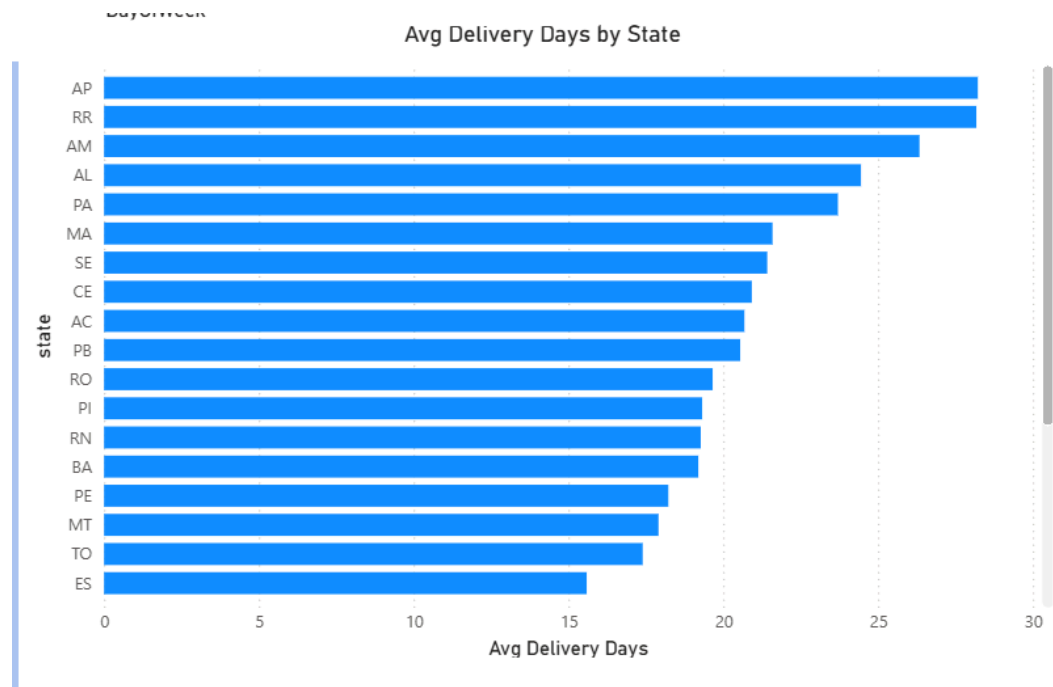
State	Avg Delivery Days
-------	-------------------

SP (São Paulo)	8.66
PR (Paraná)	11.89
MG (Minas Gerais)	11.92
ES (Espírito Santo)	12.89
SC (Santa Catarina)	14.95

Slowest States by Average Delivery Speed

State	Avg Delivery Days
AP (Amapá)	28.22
RR (Roraima)	28.17
AM (Amazonas)	26.34
AL (Alagoas)	24.45
PA (Pará)	23.7

Customers in Amapá (AP) wait on average **28.22 days**—more than **3.2 times longer** than customers in São Paulo (8.66 days). This stark bottleneck is primarily driven by complex logistics, reliance on river transport, and limited fulfillment infrastructure in Brazil's remote Northern regions.



Business Recommendation: Relying solely on Southeastern distribution hubs is unsustainable for nationwide growth. Olist must establish regional micro-fulfillment centers or partner with localized 3PL (Third-Party Logistics) providers specifically targeting the Northern cluster (**AP, RR, AM**). Bringing inventory closer to these isolated regions is critical to reducing delivery times below the 15-day threshold, which, as Insight 10 demonstrates, is the make-or-break point for preserving high customer satisfaction scores.

6.6. Customer Satisfaction & Service Quality Correlation

Insight 9: High Overall Satisfaction (4.08/5), but 1-Star Reviews Indicate Critical Bottlenecks. The average customer review score for delivered orders stands at **4.08 out of 5**. An analysis of 96,478 delivered orders reveals a strongly positive sentiment overall, with nearly 80% of customers rating their experience as 4 or 5 stars. However, the presence of over 9,300 1-star reviews signals underlying operational issues that cannot be ignored.

Review Score	Total Orders	Percentage (%)
--------------	--------------	----------------

5 Stars	56,764	58.84%
4 Stars	18,889	19.58%
3 Stars	7,908	8.20%
2 Stars	2,925	3.03%
1 Star	9,346	9.69%
<i>No Rating (Blank)</i>	<i>646</i>	<i>0.67%</i>

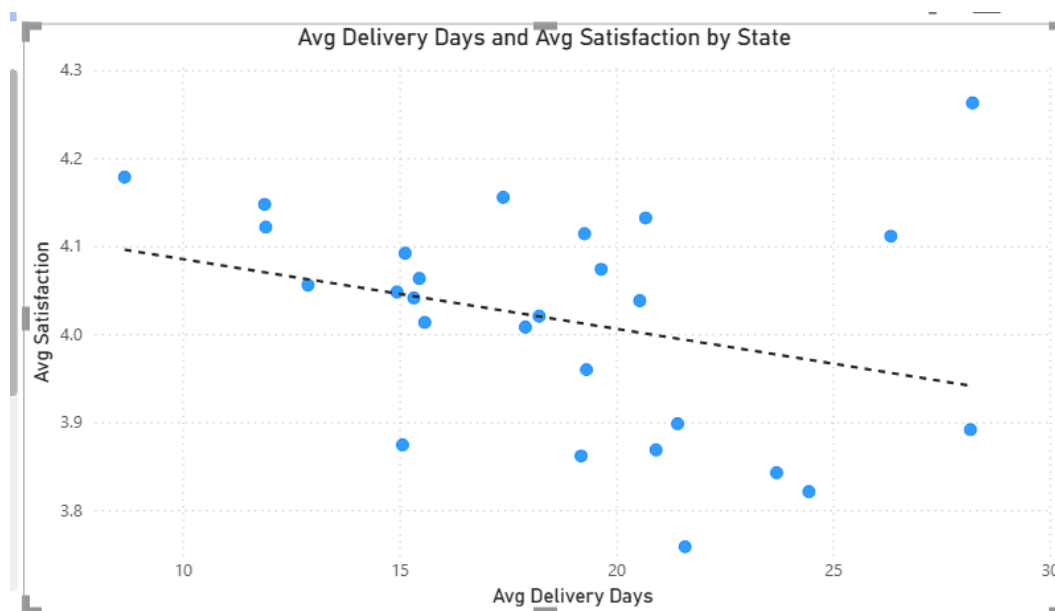
While **78.42%** of buyers are highly satisfied (combining 4 and 5-star reviews), the bottom end of the spectrum is concerning. A significant **9.69%** of transactions resulted in a 1-star review. Given the findings in previous insights, these overwhelmingly negative experiences are highly correlated with logistical failures, specifically excessively long delivery times (such as the 28+ day waits observed in Northern states like AP and RR).

Insight 10: Delivery Speed is the Strongest Predictor of Customer Satisfaction. The data reveals a perfect inverse correlation between delivery time and customer satisfaction scores. An analysis directly linking review scores to average delivery days proves that shipping speed dictates the customer's final rating:

Review Score	Avg Delivery Days
5 Stars	10.59
4 Stars	12.15
3 Stars	13.99
2 Stars	15.73
1 Star	19.54

Customers who left 5-star reviews received their orders in an average of **10.59 days**. As delivery times increase, satisfaction drops proportionally. Once the wait time stretches to an average of **19.54 days**, the transaction overwhelmingly results in a 1-star review.

This inverse correlation is visually confirmed by the scatter chart on the dashboard, where states with faster logistics (e.g., SP at 8.66 days) consistently cluster at the top of the satisfaction scale, while states enduring extended waits (e.g., AP at 28.22 days) cluster heavily at the bottom.



Business Recommendation: Implement a proactive delivery delay alert system that automatically contacts customers when an order is at risk of missing its estimated delivery window. This transparency has been shown to significantly reduce 1-star reviews even when delays occur.

PART VII: CONCLUSION

7.1. Project Summary

This Business Intelligence project successfully transformed raw, disparate e-commerce data from Olist into a structured, high-performing analytical model. By implementing a robust ETL pipeline, resolving data quality issues, and designing a Star Schema Data Warehouse, we bridged the gap between raw transactional databases and

strategic decision-making. The resulting Power BI dashboards effectively processed over 96,000 delivered orders, providing interactive, multi-dimensional insights into revenue generation, regional logistics, and customer sentiment.

7.2. Key Strategic Takeaways

Based on the data-driven insights generated by the BI dashboards, three core strategic pillars have been identified for Olist's future growth:

- **Sales & Marketing Optimization:** Revenue generation is highly dependent on seasonality (e.g., Mother's Day spikes) and early-week shopping habits (Mondays and Tuesdays). Furthermore, the dominance of credit cards and the significant reliance on 10-month installment plans highlight the necessity of flexible financing options to drive high-ticket sales.
- **Logistics & Regional Disparities:** Brazil's vast geography creates severe operational bottlenecks. While the Southeastern hub (São Paulo) drives over 38% of the revenue with high efficiency (8.66 days delivery), the Northern and Northeastern regions (e.g., Amapá, Roraima) suffer from exorbitant freight costs and unacceptable delivery times exceeding 28 days. Localized fulfillment centers are mandatory for nationwide scaling.
- **Customer Experience Management:** The data mathematically proves an undeniable inverse correlation between delivery speed and customer satisfaction. With the average 5-star experience taking 10.5 days and the average 1-star experience taking 19.5 days, maintaining an average rating above 4.0 requires a strict Service Level Agreement (SLA) that keeps delivery times under the critical 14-day threshold.

7.3. Future Scope and Enhancements

While the current BI system provides a comprehensive historical view, future iterations can be enhanced by integrating Advanced Analytics. Potential next steps include:

- **Predictive Analytics:** Integrating Machine Learning models (such as Linear Regression or Time-Series forecasting) into the data pipeline to predict future sales volumes or anticipate shipping delays before they occur.

- **Sentiment Analysis:** Utilizing Natural Language Processing (NLP) to analyze the unstructured text of customer reviews, extracting specific reasons for 1-star ratings beyond just shipping delays.
- **Real-Time Processing:** Upgrading the batch ETL process to a real-time streaming architecture to provide operational managers with live dashboards for immediate intervention.